在轉換期間從7-Mode站台的災難中恢復

ONTAP 7-Mode Transition

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在轉換期間從7-Mode站台的災難中恢復

如果您已在7-Mode主Volume與叢集Data ONTAP 式VMware次要Volume之間建立SnapMirror災難恢復（DR）關係、而且7-Mode主站台發生災難、您可以引導用戶端存取叢集Data ONTAP 式的二線實時磁碟區。當7-Mode主Volume重新連線後、您必須執行其他步驟、將用戶端重新導向至叢集Data ONTAP 式的VMware主Volume。

若要保留Data ONTAP 災難後寫入叢集式的任何資料、您必須在7-Mode主磁碟區恢復上線後、才能轉換7-Mode主磁碟區、並在叢集Data ONTAP 式的二線資料磁碟區之間建立SnapMirror關係。然後、您可以將用戶端重新導向至叢集Data ONTAP 式的等化主磁碟區。

SnapMirror不Data ONTAP 支援從叢集式的不同步到7-Mode磁碟區。因此Data ONTAP 、如果您在災難發生後重新建立7-Mode主Volume與叢集式VMware次要Volume之間的DR關係、則任何寫入次要叢集Data ONTAP 式故障恢復的資料都會遺失。

在災難發生後、將用戶端重新導向至叢集Data ONTAP 式的不二級磁碟區

如果您已在7-Mode主Volume與叢集Data ONTAP 式VMware次要Volume之間建立SnapMirror災難恢復（DR）關係、且7-Mode主站台發生災難、則必須將用戶端存取重新導向至叢集Data ONTAP 式的VMware次要Volume。

步驟

1. 從次要叢集、使用「napmirror Break」命令來中斷7-Mode主要Volume與叢集Data ONTAP 式VMware次要Volume之間的SnapMirror關係。

   ```
   sec_cluster::> snapmirror break -destination-path dst_vserver:dst_c_vol
   ```

2. 從次要叢集、使用「napmirror DELETE」命令刪除7-Mode主要Volume與叢集Data ONTAP 式的二線磁碟區之間的SnapMirror關係。

   ```
   sec_cluster::> snapmirror delete -destination-path dst_vserver:dst_c_vol
   ```

3. 將用戶端存取重新導向至叢集Data ONTAP 式的二線磁碟區。

   如需在叢集Data ONTAP 式不景環境中設定用戶端存取的詳細資訊、請參閱 "叢集Data ONTAP 式的《叢集式功能檔存取與通訊協定管理指南》"。

將7-Mode主磁碟區轉換為獨立磁碟區

災難發生後、7-Mode主Volume重新上線後、您必須轉換7-Mode主Volume。由於目前與7-Mode主磁碟區的所有SnapMirror關係都已中斷並刪除、因此您可以針對這類轉換、轉換獨立磁碟區。

步驟

1. 將資料從7-Mode磁碟區複製到叢集Data ONTAP 式的功能集區：

   a. 如果您想要設定7-Mode系統與SVM之間SnapMirror關係的TCP視窗大小、請使用「window-size-for TDP -mirror」選項建立類型為「as同步 鏡射」的SnapMirror原則。

      然後、您必須將此原則套用至7-Mode系統與SVM之間的TDP SnapMirror關係。

      您可以設定256 KB至7 MB的TCP視窗大小、以改善SnapMirror傳輸處理量、使轉換複本作業更快完成。TCP視窗大小的預設值為2 MB。

```
cluster1::> snapmirror policy create -vserver vs1 –policy tdp_policy
-window-size-for-tdp-mirror 5MB -type async-mirror
```

   b. 使用關係類型為TDP的「napmirror create」命令、在7-Mode系統和SVM之間建立SnapMirror關係。

      如果您已建立SnapMirror原則來設定TCP視窗大小、則必須將原則套用至此SnapMirror關係。

```
cluster1::> snapmirror create -source-path system7mode:dataVol20
-destination-path vs1:dst_vol -type TDP -policy tdp_policy
Operation succeeded: snapmirror create the relationship with destination
vs1:dst_vol.
```

   a. 使用「napmirror initialize」命令來啟動基礎傳輸。

```
cluster1::> snapmirror initialize -destination-path vs1:dst_vol
Operation is queued: snapmirror initialize of destination
vs1:dst_vol.
```

   b. 使用「napmirror show」命令來監控狀態。

```
cluster1::>snapmirror show -destination-path vs1:dst_vol

                            Source Path: system7mode:dataVol20
                       Destination Path: vs1:dst_vol
                      Relationship Type: TDP
                Relationship Group Type: none
                    SnapMirror Schedule: -
                 SnapMirror Policy Type: async-mirror
                      SnapMirror Policy: DPDefault
                            Tries Limit: -
                      Throttle (KB/sec): unlimited
                           **Mirror State: Snapmirrored**
                    Relationship Status: Idle
```

```
                File Restore File Count: -
                 File Restore File List: -
                      Transfer Snapshot: -
                      Snapshot Progress: -
                         Total Progress: -
              Network Compression Ratio: -
                    Snapshot Checkpoint: -
                        Newest Snapshot: vs1(4080431166)_dst_vol.1
              Newest Snapshot Timestamp: 10/16 02:49:03
                      Exported Snapshot: vs1(4080431166)_dst_vol.1
            Exported Snapshot Timestamp: 10/16 02:49:03
                                Healthy: true
                       Unhealthy Reason: -
               Constituent Relationship: false
                Destination Volume Node: cluster1-01
                        Relationship ID: 97b205a1-54ff-11e4-9f30-
005056a68289
                  Current Operation ID: -
                         Transfer Type: -
                        Transfer Error: -
                       Current Throttle: -
              Current Transfer Priority: -
                     Last Transfer Type: initialize
                    Last Transfer Error: -
                     Last Transfer Size: 152KB
Last Transfer Network Compression Ratio: 1:1
                 Last Transfer Duration: 0:0:6
                     Last Transfer From: system7mode:dataVol20
            Last Transfer End Timestamp: 10/16 02:43:53
                  Progress Last Updated: -
                Relationship Capability: 8.2 and above
                                Lag Time: -
           Number of Successful Updates: 0
               Number of Failed Updates: 0
           Number of Successful Resyncs: 0
               Number of Failed Resyncs: 0
            Number of Successful Breaks: 0
                Number of Failed Breaks: 0
                   Total Transfer Bytes: 155648
         Total Transfer Time in Seconds: 6
```

c. 視您想Data ONTAP 要手動更新叢集式的流通量或設定SnapMirror排程而定、請執行適當的行動：

如果您想要...	然後...

手動更新傳輸	i. 使用「napmirror update」命令。
	
	`cluster1::> snapmirror update -destination-path vs1:dst_vol`
	
	ii. 使用「napmirror show」命令來監控資料複製狀態。
	
	`cluster1::> snapmirror show -destination-path vs1:dst_vol`
	
	`Source Path: system7mode:dataVol20`
	
	`Destination Path: vs1:dst_vol`
	
	`Relationship Type: TDP`
	`Relationship Group Type: none`
	
	`SnapMirror Schedule: -`
	`SnapMirror Policy Type: async-mirror`
	
	`SnapMirror Policy: DPDefault`
	
	`Tries Limit: -`
	
	`Throttle (KB/sec): unlimited`
	
	`Mirror State: Snapmirrored`
	`...`
	`Number of Failed Updates: 0`
	`Number of Successful Resyncs: 0`
	`Number of Failed Resyncs: 0`
	`Number of Successful Breaks: 0`
	`Number of Failed Breaks: 0`
	`Total Transfer Bytes: 278528`
	`Total Transfer Time in Seconds: 11`

如果您想要...	然後...

執行排程的更新傳輸	i. 使用「jobschedule cron create」命令來建立更新傳輸的排程。
	
	`cluster1::> job schedule cron create -name 15_minute_sched -minute 15`
	
	ii. 使用「napmirror modify」命令、將排程套用至SnapMirror關係。
	
	`cluster1::> snapmirror modify -destination-path vs1:dst_vol -schedule 15_minute_sched`
	
	iii. 使用「napmirror show」命令來監控資料複製狀態。

2. 如果您有遞增傳輸的排程、請在準備執行轉換時執行下列步驟：
   a. 使用「napmirror quiesce」命令來停用所有未來的更新傳輸。

```
cluster1::> snapmirror quiesce -destination-path vs1:dst_vol
```

   b. 使用「napmirror modify」命令刪除SnapMirror排程。

```
cluster1::> snapmirror modify -destination-path vs1:dst_vol -schedule
""
```

   c. 如果您稍早停止SnapMirror傳輸、請使用「napmirror resume」命令來啟用SnapMirror傳輸。

```
cluster1::> snapmirror resume -destination-path vs1:dst_vol
```

3. 等待7-Mode磁碟區與叢集Data ONTAP 式VMware磁碟區之間的任何持續傳輸完成、然後中斷用戶端對7-Mode磁碟區的存取、以開始轉換。
4. 使用「napmirror update」命令、對叢集Data ONTAP 式的BIOS Volume執行最終資料更新。

```
cluster1::> snapmirror update -destination-path vs1:dst_vol
Operation is queued: snapmirror update of destination vs1:dst_vol.
```

5. 使用「shnapmirror show」命令來驗證上次傳輸是否成功。
6. 使用「napmirror Break」命令來中斷7-Mode Volume與叢集Data ONTAP 式VMware Volume之間的SnapMirror關係。

```
cluster1::> snapmirror break -destination-path vs1:dst_vol
[Job 60] Job succeeded: SnapMirror Break Succeeded
```

7. 如果您的磁碟區已設定LUN、請在進階權限層級使用「LUN Transition 7-mode show」命令來驗證LUN是否已轉換。

   您也可以在叢集Data ONTAP 式的支援區上使用「LUN show」命令來檢視所有已成功轉換的LUN。

8. 使用「napmirror DELETE」命令刪除7-Mode Volume與叢集Data ONTAP 式VMware Volume之間的SnapMirror關係。

```
cluster1::> snapmirror delete -destination-path vs1:dst_vol
```

9. 使用「napmirror release」命令、從7-Mode系統移除SnapMirror關係資訊。

```
system7mode> snapmirror release dataVol20 vs1:dst_vol
```

將用戶端重新導向至叢集Data ONTAP 式的功能區

當7-Mode主Volume恢復上線後、您可以轉換7-Mode主Volume、與叢集Data ONTAP 式的SnapMirror次要Volume建立SnapMirror關係、並將用戶端存取重新導向至叢集Data ONTAP 式的VMware主Volume。

步驟

1. 在主要和次要SVM之間建立SVM對等關係。
   a. 使用「cluster對等create」命令建立叢集對等關係。

   ```
   pri_cluster::> cluster peer create -peer-addrs cluster2-d2,
   10.98.234.246 -timeout 60

   Notice: Choose a passphrase of 8 or more characters. To ensure the
   authenticity of the peering relationship, use a phrase or sequence of
   characters that would be hard to guess.

   Enter the passphrase: ********
   Confirm the passphrase: ********
   ```

   b. 從來源叢集、使用「vserver對等create」命令、在叢集Data ONTAP 式的實體化主磁碟區和叢集Data ONTAP 式的二線實體磁碟區之間建立一個SVM對等關係。

   ```
   pri_cluster::> vserver peer create -vserver src_vserver -peervserver
   src_c_vserver -applications snapmirror -peer-cluster sec_cluster
   ```

   c. 從目的地叢集、使用「vserver對等端點接受」命令來接受SVM對等端點要求、並建立SVM對等端點關係。

   ```
   sec_cluster::> vserver peer accept -vserver dst_vserver -peervserver
   src_vserver
   ```

2. 使用「napmirror create」命令建立SnapMirror關係、將叢集Data ONTAP 式的等二級磁碟區作為來源、並Data ONTAP 將叢集式的等一級磁碟區作為目的地。

   ```
   pri_cluster::> snapmirror create -source-path dst_vserver:dst_c_vol
   -destination-path src_vserver:src_c_vol
   ```

3. 從主叢集、使用「napmirror resSync」命令重新同步叢集Data ONTAP 式的二線Volume。

```
pri_cluster::> snapmirror resync -source-path dst_vserver:dst_c_vol
-destination-path src_vserver:src_c_vol
```

   您必須等到重新同步完成。SnapMirror狀態會在重新同步完成時變更為「naprror」（快照複本）。

4. 當您準備好切換至叢集Data ONTAP 式的「叢集式」功能主磁碟區時、請中斷用戶端對叢集Data ONTAP 式「二線」磁碟區的存取。
5. 從主叢集、使用「napmirror update」命令來更新主磁碟區。

```
pri_cluster::> snapmirror update -destination-path src_vserver:src_c_vol
```

6. 從主叢集、使用「napmirror Break」命令來中斷叢集Data ONTAP 式故障主磁碟區和次磁碟區之間的SnapMirror關係。

```
pri_cluster::> snapmirror break -destination-path src_vserver:src_c_vol
```

7. 讓用戶端能夠存取叢集Data ONTAP 式的等化主磁碟區。
8. 從主叢集中、使用「napmirror DELETE」命令刪除叢集Data ONTAP 式的SnapMirror主要和次要磁碟區之間的SnapMirror關係。

```
pri_cluster::> snapmirror delete -destination-path src_vserver:src_c_vol
```

9. 從次要叢集、使用「napmirror cred」命令建立SnapMirror關係、將叢集Data ONTAP 式的等一級磁碟區做為來源、並Data ONTAP 將叢集式的等二級磁碟區做為目的地、 排程類似於先前排程的7-Mode主Volume與叢集Data ONTAP 式的VMware次要Volume。

```
sec_cluster::> snapmirror create -source-path src_vserver:src_c_vol
-destination-path dst_vserver:dst_c_vol -schedule 15_minute_sched
```

10. 從次要叢集、使用「napmirror resSync」命令重新同步叢集Data ONTAP 式的主Volume。

```
sec_cluster::> snapmirror resync -source-path src_vserver:src_c_vol
-destination-path dst_vserver:dst_c_vol
```

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